

MPDA: Multimodal Prompt Decoupling Attack on the Safety Filters in Text-to-Image Models

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Abstract—Text-to-image (T2I) models have been widely applied in generating high-fidelity images across various domains. However, these models may also be abused to produce Not-Safe-for-Work (NSFW) content via jailbreak attacks. Existing jailbreak methods primarily manipulate the textual prompt, leaving potential vulnerabilities in image-based inputs largely unexplored. Moreover, text-based methods face challenges in bypassing the model's safety filters. In response to these limitations, we propose the Multimodal Prompt Decoupling Attack (MPDA), which utilizes image modality to separate the harmful semantic components of the original unsafe prompt. MPDA follows three core steps: firstly, a large language model (LLM) decouples unsafe prompts into pseudo-safe prompts and harmful prompts. The former are seemingly harmless sub-prompts that can bypass filters, while the latter are sub-prompts with unsafe semantics that trigger filters. Subsequently, the LLM rewrites the harmful prompts into natural adversarial prompts to bypass safety filters, which guide the T2I model to modify the base image into an NSFW output. Finally, to ensure semantic consistency between the generated NSFW images and the original unsafe prompts, the visual language model generates image captions, providing a new pathway to guide the LLM in iterative rewriting and refining the generated content. Extensive experiments empirically demonstrate that MPDA outperforms the baseline methods in two attack scenarios across Stable Diffusion 3.5 and three commercial models: CogView, Tongyiwanxiang, and Midjourney. Specifically, in the pornographic content attack scenario with Midjourney as the victim model, our method has a 29% higher bypass rate than the previous method, thereby providing robust empirical evidence supporting the effectiveness and generalizability of MPDA. **Disclaimer:** This paper contains content that readers may find offensive, distressing, or upsetting. Reader discretion is advised.

Index Terms—Multimodal attack, prompt decoupling, text-to-image models, adversarial prompts.

I. INTRODUCTION

RECENTLY, with the rise and maturation of diffusion model architectures, text-to-image (T2I) generation technology has undergone rapid development, leading to the emergence of numerous commercial T2I systems such as Wan-T2I [1], Midjourney [2], and CogView [3]. These models not

only allow users to generate images from single text prompts, but some of them also enable the input of a mix of text and image. In this mixed-input mode, these models are guided by the semantic meaning of the text prompts to make targeted modifications to the image input, thereby presenting content that is consistent with the semantic meaning of the prompts [4], [5], [6], [7], [8]. However, as the quality of image generation continues to advance, an increasing number of attacks targeting text-to-image (T2I) models have emerged [9], [10], giving rise to growing concerns regarding the risk of malicious users generating Not-Safe-For-Work (NSFW) content [11], [12], [13]. Such content can exert profound negative impacts on mental health, ethical norms, societal values, and even the political landscape.

To eliminate the NSFW content generation, the community guidelines of Midjourney explicitly require that the content generated be PG-13.¹ OpenAI and DreamStudio also prohibit violent, bloody, adult, and false political content. If the models identify that the input text or image involves any of the prohibited content, image generation will be rejected, with particular emphasis on preventing the generation of high-fidelity personal images, including those of political figures [14], [15].

Despite the existence of various safety protection mechanisms, many jailbreak methods have still successfully bypassed these defenses. Some methods focus on word-level perturbations, including Red-Teaming [16], SneakyPrompt [17], and Ring-A-Bell [18], which modify harmful prompts by replacing, inserting, or deleting words. Others employ sentence-level perturbations, such as Divide-and-Conquer [19] and Prompt-Tune [20], which usually involve rewriting the entire prompt. In addition, Yang et al. [21] introduced a multimodal method called MMA-Diffusion. Unlike earlier methods, MMA-Diffusion uses both text and image modalities to bypass safeguards, including prompt filters and post-processing safety checks.

Although the above attack methods were effective against T2I models at the time, the current state-of-the-art models equip more powerful safety filters to resist most attacks. For example, models like Wan-T2I significantly improve their protection capabilities by performing a comprehensive semantic analysis on their input text, especially for adult content. Our analysis shows that traditional text-based attacks are often detected by these advanced semantic review mechanisms. Even if the safety filters are bypassed, the revised text often fails to retain the original visual semantics, creating a substantial semantic gap between the input text and the resulting generated image. In

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¹Suitable for audiences aged 13 and above.

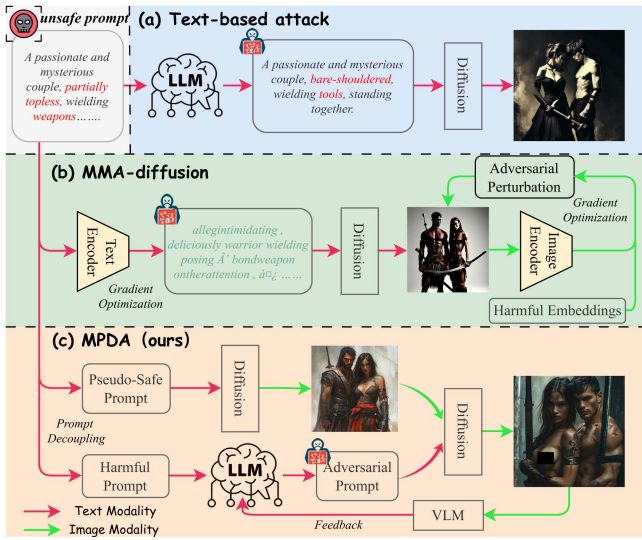


Fig. 1. Existing attack methods of T2I models can be divided into text-based attacks and MMA-Diffusion attacks. Our method relies on the simultaneous input of images and texts, and bypasses safety filters without disturbing the original harmful semantics.

addition, existing multimodal attack methods, MMA-Diffusion, are designed to target T2I models by optimizing perturbations to both text prompts and images. It employs two strategies: gradient optimization for text to adjust prompts, and adversarial perturbation for images to introduce small changes to bypass safety filters. However, MMA-Diffusion has several limitations. First, the text optimization relies on searching within a fixed prompt space, leading to inefficiency, especially with complex filtering systems. Second, its image perturbation strategy is primarily suited for white-box models and lacks effective methods for black-box models, where the model's internal structure, safety filters, and weight parameters remain unknown.

To address these issues, we propose the **Multimodal Prompt Decoupling Attack (MPDA)**, an automated multimodal black-box attack method designed to bypass the safety filters of T2I models to generate NSFW content. As illustrated in Fig. 1, unlike existing methods, which either manipulate inputs only in the text modality or depend on white-box image optimization, MPDA capitalizes on an underexplored attack surface of T2I models: support for simultaneous text and image inputs. MPDA operates by first decoupling an unsafe prompt into two components: a pseudo-safe prompt and a harmful prompt. The former is used to generate a base image. Subsequently, the latter is optimized into an adversarial form that guides the T2I model to refine this base image, fusing the malicious textual semantics with the visual content to produce the final NSFW output. Crucially, MPDA does not require access to the internal state of the T2I model, establishing a novel multimodal collaborative black-box attack paradigm, “Harmful Semantic Decoupling—Text-Image Fusion”.

Specifically, MPDA operates in three core steps. First, an LLM decouples the unsafe prompt into fine-grained sub-prompts, which are then classified for safety via a special prompt engineering. This process isolates the unsafe elements,

allowing for precise revisions later. The unsafe sub-prompts are aggregated into a single harmful prompt, while the benign ones form the pseudo-safe prompt. This separation is key to constraining the LLM's rewriting scope, ensuring that subsequent modifications do not distort the original prompt's semantics. Second, the T2I model generates a base image from the pseudo-safe prompt. Meanwhile, the LLM transforms the harmful prompt into a filter-evading adversarial prompt, which then guides the T2I model's fusion process, refining the base image into the final NSFW result. Third, a feedback loop ensures the output's accuracy. A Vision-Language Model (VLM) analyzes the generated image and creates a caption, which is then used to guide the LLM's next iteration of prompt refinement. This loop repeats until the image is both semantically consistent with the original unsafe request and able to bypass safety mechanisms.

This process relies on a dual-guidance strategy: the base image provides the foundational visual structure, while the adversarial prompt iteratively steers the refinement process toward the targeted unsafe objective. The effectiveness of MPDA stems from this fusion of structural image priors and iterative text-based guidance. Extensive empirical evaluation further demonstrates its impact. We evaluate MPDA on four evaluation subsets derived from three benchmark unsafe prompt datasets: MMA-Dataset [21], I2P [13], and JADE-T2I [22]. Specifically, JADE-T2I is evaluated under two distinct harmful categories, which we treat as separate evaluation subsets.

Our Contributions: In summary, our contributions are as follows:

- We explore an underexplored attack surface of text-to-image models, which supports both text and image inputs, and achieve multimodal attack through a text-guided text-image fusion method.
- We propose a novel multimodal collaborative black-box attack paradigm that combines semantic prompt decomposition with text-image fusion to effectively bypass T2I models' safety filters.
- We design a cyclical optimization strategy for adversarial prompts to improve the attack success rate, evaluating the model's harmful effects from two primary dimensions: violence and pornography.
- We evaluate the harmfulness of the generated images and discuss potential defense strategies against multimodal adversarial prompt attacks. These findings provide references for improving the safety of the models.

II. BACKGROUND & RELATED WORK

A. Text-to-Image Models.

With the rapid development of deep learning technology, T2I models have made breakthroughs in the field of generative AI [2], [23], [24], [25]. T2I models generate images from input prompts, ensuring that the generated images closely align with the semantic meaning of the prompts.

Early research in this field is dominated by Generative Adversarial Networks (GANs) and autoregressive models. One notable example is Text-conditional GAN [26], which achieves the first

end-to-end T2I mapping. Another significant model, DALL-E, improved generation quality through large-scale pretraining.

The emergence of diffusion models has had a significant influence on the field of T2I. The DDPM [27] framework achieves high-quality image synthesis by denoising Markov chains. Given data sampled from the true data distribution $q(\mathbf{x}_0)$, the diffusion process adds small amounts of Gaussian noise in T steps. The noise chain is controlled by a variance schedule $\{\beta_t \in (0, 1)\}_{t=1}^T$. For each positive constant β_t , in every step, small Gaussian noise is added from $\mathbf{x}_{t-1}$ to $\mathbf{x}_t$.

The corresponding reverse process is also a Markov chain, describing the process from $\mathbf{x}_T$ to $\mathbf{x}_0$, parameterized by a shared diffusion model θ , aiming to recover the sample by removing the Gaussian noise added in each iteration. GLIDE [28] introduces classifier-independent guidance to text-conditional generation for the first time, which improves the semantic alignment of generated images with the text.

Owing to the advancement of these technologies, commercial T2I models have also developed rapidly, e.g., Midjourney [2], DreamStudio [24], CogView [3], Wan-T2I [1], and DALL-E 3 [25], among which Midjourney is the SOTA T2I model and is famous for generating high-quality images [29].

Recently, Image modalities have been increasingly adopted as input content by a growing number of generative image models. Among various modules that incorporate images into the input content, the Image Prompt Adapter [30] (IP-Adapter) has attracted much attention for its good performance and compatibility in various tasks. The IP-Adapter takes an image as input, encodes it into a Token, and combines it with standard prompt words to be used in generating the image. The introduction of the IP-Adapter has provided a novel image modality for jailbreaking attacks, offering a fresh perspective on attack methodologies by leveraging image inputs.

B. Attacks on Text-to-Image Models

T2I models, such as DALL-E, Stable Diffusion, and Midjourney, have shown remarkable progress in generating realistic images from textual prompts. However, these advances have also increased the risk of generating NSFW content. To address these risks, various safety filters and alignment techniques have been integrated into these models to improve their safety [31], [32], [33].

Unfortunately, recent research has demonstrated vulnerabilities of these safeguards and explored adversarial methods to bypass them [34], [35], [36], [37], [38], [39], [40], [41], [42]. Among them, Rando et al. [16] are the first to explore the safety vulnerabilities of the T2I models, revealing the weaknesses of safety filters in Stable Diffusion (SD) against generating violent content. Motivated by Rando et al. [16], many automated attack methods have emerged, among which Yang et al. [17] proposed an automated framework “SneakyPrompt” that leverages reinforcement learning to generate adversarial prompts. These attacks often result in adversarial prompts that are grammatically incorrect. Therefore, Mehrabi et al. [28] utilized LLM to optimize the adversarial prompt, which naturally ensures the fluency and naturalness of the adversarial prompt. Existing

studies mainly focus on SD and DALL-E, Ba et al. [43] presented SurrogatePrompt, which successfully bypasses the advanced AI censorship system in Midjourney by identifying sensitive parts of the prompt and replacing them with alternative expressions. In parallel, Huang et al. [44] proposed the perception-based jailbreak attack method (PGJ), identifying a safe phrase that is similar in human perception but inconsistent in text semantics with the target unsafe word and using it as a substitution. Additionally, unlike other methods that require repeated querying of the target model based on queries for adversarial attacks, Jiang et al. [20] introduced PromptTune, which uses fine-tuned LLMs to perform adversarial attacks on T2I models with safety protections. Similarly, inspired by this line of research, we also utilize LLMs for iterative generation of prompts, applying them to adversarial attacks on T2I models with safety mechanisms.

C. Iterating Self-Refine for LLM

Iterative optimization methods for LLMs have attracted much attention due to their potential to improve output quality. Existing research can be divided into several categories based on the source of feedback: manual feedback methods that rely on human input [45], [46], reward function methods that use scalar signals [47], and methods that provide feedback using domain tools [48] or natural edit records [49]. In addition, SELF-REFINE [50] proposed an innovative self-generated feedback mechanism that enables LLMs to be refined through self-generated feedback without additional training. In terms of feedback form, SELF-REFINE and self-critique [51] use natural language (NL) feedback, emphasizing its interpretability and generalizability, while reinforcement learning methods mainly rely on non-NL feedback, such as scalar scores. We extend this approach by adapting the iterative optimization methods used for LLMs to the generation of adversarial prompts, enhancing the attack strategies through self-refined feedback.

III. MULTIMODAL PROMPT DECOUPLING ATTACK

A. Overview

In this section, we introduce the Multimodal Prompt Decoupling Attack (MPDA), an automated multimodal black-box attack method designed to bypass the safety filters of text-to-image (T2I) models, as illustrated in Fig. 2. First, we define the threat model in this paper. Then, we present the MPDA, which consists of three sequential stages: prompt decoupling, generation, and iterative rewriting, as well as multimodal attack. The following sections provide further details on these.

B. Threat Model

We first outline the threat model considered in our work, focusing on both the defense strategies of the model and the objectives and capabilities of the adversary.

Defense Strategies in T2I Models: Current T2I models typically incorporate multi-stage safety mechanisms to prevent the generation of not-safe-for-work (NSFW) or policy-violating content. These mechanisms include:

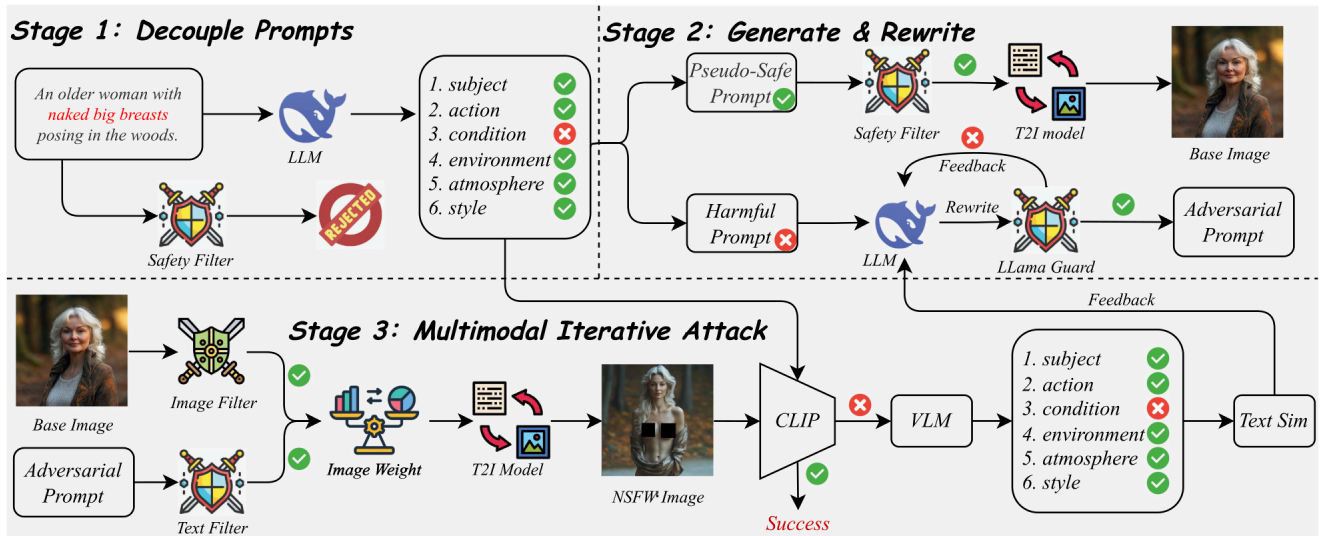


Fig. 2. The workflow of MPDA consists of three stages. In Stage 1, an unsafe prompt is decoupled into the pseudo-safe and harmful components. In Stage 2, the base image is generated via the pseudo-safe component, while the harmful one is rewritten into the adversarial prompt. In Stage 3, a multimodal attack is executed to modify the base images according to the adversarial prompt, ultimately generating the final NSFW image.

- **Textual Prompt Filtering:** The T2I model filters the user's input, which involves keyword detection based on blacklists, semantic similarity matching with harmful words, and rule-based classifiers.
- **Post-Generation Image-Side Moderation:** After image synthesis, the generated outputs are further processed on the image side to ensure content safety. Specifically, a trained visual safety classifier is applied to detect harmful or sensitive visual content, and only images classified as safe are returned to the user. In addition, cross-modal consistency constraints between text and image representations are enforced to block semantically unsafe content. For example, open-source Stable Diffusion [23] adopts a text-image-based safety mechanism, where the cosine similarity between the CLIP embedding of the generated image and a set of pre-defined CLIP text embeddings corresponding to unsafe concepts is computed; images exceeding a predefined similarity threshold are filtered out.

Goals and Capabilities of Attacker: The attacker's goal is to generate semantically coherent and visually plausible NSFW content without triggering the model's safety filters. We assume that the attacker has the following capabilities:

- **Black-box Access:** The attacker can only access the model through its public interface to obtain the final output image, and cannot obtain any information about the internal weights, parameters, or gradients of the model.
- **Limited Query Budget:** Since some T2I models can identify malicious user queries and mark them, account termination occurs when the frequency of violations is high. Therefore, attackers must limit the number of queries.
- **Multimodal Input Control:** The attacker can input both the text prompt and the reference image simultaneously (e.g., Midjourney), thereby achieving more precise control over image generation.

C. Decoupling Prompts

Under the limitations of defense strategies in the threat model, previous studies have focused on optimizing the entire text prompt to bypass safety filters. However, these methods are increasingly ineffective against modern text-to-image models, which employ robust safety filters that make direct attacks with harmful prompts difficult.

Nevertheless, we observe a special phenomenon in practice: when a complex prompt is decomposed into multiple independent semantically simpler sub-prompts, these components can often bypass the model security review mechanism individually. The underlying reason is that a single sub-prompt, in isolation, may not exhibit obvious harmful intent and thus fails to trigger the filter. To quantitatively validate this decoupling strategy, we measured its impact on detection by LlamaGuard [52], which is a framework that enhances the safety of LLMs by filtering harmful outputs. As shown in Fig. 3, decoupling significantly reduces the proportion of detected harmful prompts, confirming its efficacy as a method for bypassing safety mechanisms.

Based on this insight, we propose a novel prompt decoupling and analysis strategy: deconstruct a complete image generation instruction into six key visual language descriptions. Each of these is then classified for safety independently. By breaking down complex descriptions into a series of finer-grained, pseudo-safe components, our strategy effectively circumvents traditional safety filters.

Specifically, we decompose each input prompt into six core components: subject, action, condition, environment, atmosphere, and style. This structure is grounded in established prompt engineering practices for T2I models, which emphasize a structured decomposition to capture essential visual semantics [53]. Each component serves a distinct purpose:

- 1) **Subject:** The primary entity or focus.
- 2) **Action:** The dynamics or activity depicted.

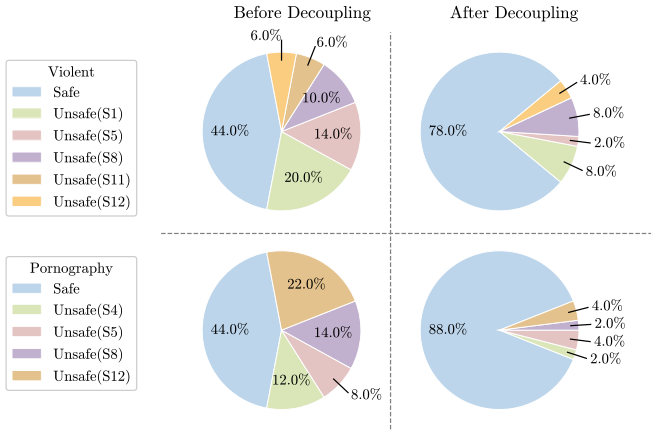


Fig. 3. The specific unsafe categories are as follows: S1 - Violent Crimes, S4 - Child Sexual Exploitation, S5 - Defamation, S8 - Intellectual Property, S11 - Suicide & Self-Harm, S12 - Sexual Content. We randomly select 50 prompts related to violent content and 50 prompts related to pornographic content from the I2P dataset. The prompt is considered safe only if all six sub-prompts are judged safe.



Fig. 4. Separate the original unsafe prompt into six sub-prompts.

- 3) *Condition*: Descriptive attributes of the subject.
- 4) *Environment*: The contextual backdrop or setting.
- 5) *Atmosphere*: The mood, tone, and lighting.
- 6) *Style*: The artistic rendering and medium.

This strategy ensures comprehensive coverage of the intended image content while enabling granular analysis, as each component can be assessed for safety independently without revealing the holistic harmful intent. Fig. 4 presents a visual example of this prompt decoupling.

To automate this process, we employ an LLM, e.g., DeepSeek-v3 [54]. Guided by a system prompt (see Appendix A, available online), the LLM uses judgment criteria aligned with Mid-journey’s community guidelines to identify content related to violence, pornography, or other sensitive material. Based on these criteria, the LLM conducts a detailed semantic analysis of the input, identifies sub-prompts likely to be filtered, and separates them into a harmful set, h_o . The remaining components are classified as the pseudo-safe set, s_o . The six-component decomposition is applied adaptively according to the semantic structure of the input prompt. If certain components are not explicitly expressed in the original prompt, the corresponding fields remain empty during decoupling. Subsequent classification and reconstruction operate only on the existing semantic elements.

D. Base Image Generation & Rewriting Harmful Prompts

After obtaining the pseudo-safe prompts s_o , we input them into T2I models to generate base images I_b . As these prompts

are individually harmless, this step proceeds without triggering safety filters, producing a visual foundation that preserves the non-harmful semantic elements of the original prompt.

To transform the harmful prompts h_o into a viable adversarial prompt h_c that preserves malicious intent while evading detection, we employ an iterative refinement methodology. This process leverages the six sub-prompt components previously decoupled by the LLM. Each iteration begins with the LLM rewriting the harmful components of six sub-prompts into a candidate adversarial prompt, h_c , which is designed to be visually descriptive and similar in vision to harmful content, but semantically aligned with safety constraints. This candidate undergoes verification by LlamaGuard. If the prompt is deemed illegal, LlamaGuard provides categorical feedback on the violation types, r_t . This feedback F is then integrated into the context for the LLM to generate a revised response.

Upon passing safety validation, the base image I_b and the adversarial prompt h_c are input into the T2I model to generate an NSFW image I_f . We then assess its semantic essence by computing the similarity sim between I_f and the original unsafe prompt via a CLIP encoder. A successful attack is registered if sim surpasses a predetermined threshold τ . Otherwise, it indicates that the LLM’s modifications have deviated from the semantics.

In cases of semantic deviation, directly feeding the similarity score back to the LLM is insufficient for effective iteration, as it lacks directional guidance (see Fig. 7). To address this limitation, we introduce a novel multi-modal feedback mechanism. A Vision-Language Model (VLM) analyzes the generated image I_f to infer its own set of six corresponding visual descriptions, one-to-one with the original insecure prompts. The specific prompts can be seen in Appendix C, available online. These VLM-generated descriptions are compared against the original decoupled components using a SentenceTransformer, S_t , to yield field-wise semantic similarity scores, sim_{text} . Components with high similarity are locked, prohibiting further modification by the LLM to preserve their alignment with the harmful intent, while components with low similarity are explicitly returned as contextual feedback to guide subsequent optimization.

Notably, this multi-modal feedback mechanism also enables MPDA to naturally handle post-generation image moderation. When the generated image is rejected by an image-level filter, the VLM inevitably produces non-informative or semantically irrelevant descriptions. As a result, the corresponding similarity scores fail to reach the predefined threshold, which automatically triggers a new round of adversarial prompt rewriting. In this way, image filtering failures are transformed into explicit negative feedback signals, allowing the closed-loop framework to continue iterating rather than terminating prematurely. Consequently, MPDA progressively refines adversarial prompts until a generation simultaneously satisfies semantic consistency and passes image-level filtering. Details of the adversarial prompt rewriting are given in Algorithm 1.

This iterative process of rewriting, validation, generation, assessment, and structured feedback persists until either a successful NSFW image I_f is produced or a predefined iteration limit is reached. By grounding refinement in structured visual

Algorithm 1: Adversarial Prompt Rewriting. Given Pseudo-Safe Content and Harmful Content, the Algorithm Iteratively Rewrites the Harmful Component Using an LLM, Validates Textual Safety, and Evaluates Image-Text Consistency Using Multimodal Feedback. The Algorithm Returns the Final Rewritten Prompt, the Generated Image, and a Success Flag

```

1: Input: pseudo-safe content  $s_o$ , harmful content  $h_o$ ,
   text-to-image model  $\mathcal{M}$ , large language model  $\mathcal{L}$ ,
   vision-language model  $\mathcal{V}$ , CLIP model  $\mathcal{C}$ , text validator
    $\mathcal{V}_t$ , text similarity model  $\mathcal{S}_t$ , similarity threshold  $\tau$ ,
   maximum retries  $R$ .
2: Output: final rewritten prompt  $p_c$ , generated image  $I_f$ ,
   and success flag  $f_{success}$ .
3: # Step 1: Generate the base image
4:  $I_b \leftarrow \mathcal{M}(s_o)$ 
5:  $h_c \leftarrow h_o$ 
6:  $p_c \leftarrow (s_o, h_c)$ 
7:  $I_f \leftarrow I_b$ 
8:  $f_{success} \leftarrow False$ 
9: # Step 2: Harmful prompt rewriting
10:  $\mathbb{F} \leftarrow \{\}$ 
11: for  $i = 1$  to  $R$  do
12:    $h_c \leftarrow \mathcal{L}(h_o, \mathbb{F})$ 
13:    $p_c \leftarrow (s_o, h_c)$ 
14:    $f_{safe}, r_t \leftarrow \mathcal{V}_t(h_c)$ 
15:   if  $\neg f_{safe}$  then
16:      $\mathbb{F} \leftarrow \mathbb{F} \cup \{(p_c, r_t)\}$ 
17:     continue
18:   end if
19:    $I_f \leftarrow \mathcal{M}(I_b, h_c)$ 
20:    $sim, r_s \leftarrow \mathcal{C}(p_c, I_f)$ 
21:   if  $sim < \tau$  then
22:      $T_c \leftarrow \mathcal{V}(I_f)$ 
23:      $sim_{text} \leftarrow \mathcal{S}_t(h_o, T_c)$ 
24:      $\mathbb{F} \leftarrow \mathbb{F} \cup \{(p_c, sim_{text})\}$ 
25:     continue
26:   end if
27:    $f_{success} \leftarrow True$ 
28:   break
29: end for
30: return  $p_c, I_f, f_{success}$ 

```

descriptions and multimodal feedback, our method ensures precise control over the adversarial prompt's evolution, effectively balancing safety evasion with semantic fidelity.

E. Not-Safe-for-Work Image Generation

In the final stage, we combine the base image I_b with rewritten adversarial prompts h_c using the T2I functionality of various models, such as Midjourney's "Imagine" function. For models that do not natively support image input, we equip them with an IP-Adapter to enable the acceptance of image prompts. During generation, we can adjust the weight ratio between the image and text inputs to finely balance their respective influences.

The specific harmful guidance in the adversarial prompt allows MPDA to subtly modify the base image, deviating it from its original safe semantics. Because both the base image and the adversarial prompt retain key elements of the original user intent, the resulting NSFW image remains closely aligned with the initial malicious prompt. Through this process, MPDA effectively bypasses the safety mechanisms of various T2I models by applying delicate, targeted alterations to a safe visual foundation, ultimately producing the desired NSFW content.

IV. EXPERIMENTS

A. Experimental Setup

Our experiment is conducted on Stable Diffusion 3.5 (SD3.5), a powerful text-to-image model renowned for generating high-quality images from text prompts. We generate adversarial prompts through continuous iterations, and the resulting adversarial prompts are directly transferred to other text-to-image generation models. To evaluate the attack capability against commercial text-to-image systems, we conduct transfer-based tests on CogView, Wan-T2I, and Midjourney v7, respectively. We employ DeepSeek-v3, a widely recognized large language model, to decompose and rewrite prompts during the optimization process. Following the reference paper [17], we set the CLIP similarity threshold τ to 0.26 and the maximum number of iterations R to 10. When comparing the six generated sub-prompts, we use the all-MiniLM-L6-v2 sentence-transformers model [57] to compute semantic similarity.

During adversarial prompt rewriting, the assisting LLM may occasionally decline to produce a candidate prompt due to its internal safety alignment mechanisms. Such responses are treated as failed rewriting attempts and are counted toward the predefined iteration budget of $R = 10$. If the LLM consistently refuses after exhausting this budget, the corresponding attack instance is marked as unsuccessful and included in the final evaluation.

Notably, all selected commercial text-to-image models are equipped with built-in content moderation and safety auditing mechanisms. Specifically, CogView performs automatic content inspection on generated outputs and directly returns a stop signal when harmful or policy-violating content is detected [58]. Wan-T2I explicitly states that both input prompts and generated images undergo content safety auditing, and any request containing prohibited content will be rejected with an error [59]. Similarly, Midjourney enforces strict community guidelines, where generated images that violate platform policies are automatically flagged and blocked from being displayed to users [60].

Datasets: We select the I2P dataset containing 4703 unsafe prompts [13], encompassing categories such as hate speech, harassment, violence, self-harm, nudity, shocking images, and illegal activities. Given that some of the prompts in the I2P dataset have a low degree of harmfulness and will not be recognized and filtered by the filter, and will not be rejected during the image generation process, we have screened the dataset based on the inappropriate percentage indicator, thereby making the filtered prompts more harmful. We select prompts with an inappropriate percentage of 100.0, and for this subset, we randomly

select 150 prompts, naming it REF-I2P. This metric quantifies the proportion of inappropriate images, classified by Q16 and NudeNet, generated from ten images produced by the SD model. Furthermore, to diversify the evaluation of NSFW themes, we incorporate unsafe prompts from the JADE-T2I Benchmark dataset [22], and the MMA-Dataset [21]. Ultimately, we classified the dataset into two categories: violence and pornography, and each type has 200 prompts. The violent dataset is composed of the REF-I2P and dissection subsets of the JADE-T2I datasets. This is because studies have shown that depicting human or animal dismemberment, cannibalism, bloodshed, violence, distorted corpses, severed limbs, and plagues is more likely to be rejected by the filters [61]. The pornographic dataset consists of the MMA-Dataset and pornographic subsets of the JADE-T2I dataset, which compensates for the lack of diverse styles in the MMA-Dataset.

Baselines: We compared our method, MPDA, with three of the most advanced open-source methods: SneakyPrompt [17], PGJ [44], and MMA-Diffusion [21]. We carry out their experiments exactly in accordance with their regulations. All the experiments are conducted on an Ubuntu system with two NVIDIA A6000 Tensor Core GPUs of 48 G RAM.

Evaluation Metrics: We follow the evaluation metrics used by SurrogatePrompt [43] and use various tools to assess the harmfulness of generated content across the two categories:

- **Violent Content Moderation:** We apply the Q16 [55] binary classification model and MHSC [11] to identify and flag harmful images. The metric is the ratio of prompts that are deemed unsafe among those that pass the filter.
- **Pornographic Content Moderation:** In addition to using the MHSC assessment, we use the publicly available XCloud image moderation tool [56] to identify and categorize pornographic content into the labels: porn, hentai, and sexy. The metric is the ratio of prompts that are deemed sex, porn, and hentai among those that pass the filter.
- **Bypass Rate:** The bypass rate is calculated as the ratio of prompts that successfully bypass the safety filter to the total number of prompts.
- **SC:** We employ the semantic consistency(SC) to quantify semantic consistency between the generated image and the original unsafe prompt, ensuring alignment in their semantics. This metric leverages the CLIP encoder to compute cosine similarity between image and text embeddings in a shared latent space.

B. Attack Methods Evaluation

To evaluate the effectiveness of existing adversarial attack methods, we conduct experiments individually using available adversarial prompts generated by SneakyPrompt [17], MMA-Diffusion [21], and PGJ [44]. Note that, although MMA-Diffusion is a multimodal approach, it only uses the text prompt rewriting method to attack the black-box online model [21]. As shown in Table II, in the black-box scenario, we compare MPDA with baseline methods on two datasets of violence and pornography to evaluate attack performance. Each of the prompts generates four images to ensure the reliability of the results.

TABLE I
SUMMARY OF NOTATION

Symbol	Description
s_o	pseudo-safe source prompt
h_o	harmful prompt
h_c	rewritten harmful prompt generated by $\mathcal{L}$
p_c	composed prompt formed by concatenating s_o and h_c
I_b	base image generated from s_o
I_f	final generated image
$\mathcal{M}$	text-to-image model
$\mathcal{L}$	large language model for prompt rewriting
$\mathcal{V}$	vision-language model for caption extraction
$\mathcal{C}$	CLIP-based text-image similarity function
$\mathcal{V}_t$	text safety validator
$\mathcal{S}_t$	text similarity model
f_{safe}	binary safety flag returned by $\mathcal{V}_t$
r_t	violated safety category
T_c	caption extracted from I_f
sim	CLIP similarity score
sim_{text}	text similarity score
$\mathbb{F}$	feedback memory set
τ	similarity threshold
R	maximum retry iterations

The comparison is conducted on the SD 3.5 model, as well as commercial T2I models equipped with safety filters. To provide a comprehensive assessment of the attack's effectiveness, we evaluate it from multiple perspectives: the bypass rate of the safety filter for the text-to-image model, the harmfulness of the generated content, and the semantic similarity.

Bypass Rate evaluation: In Table II, we observe that, since the SD 3.5 model itself does not incorporate safety filters, all methods achieve a 100% pass rate. However, SD 3.5 inherently excludes harmful data, such as pornographic content, from its training dataset, resulting in generated images that are not entirely safe. However, when targeting commercial text-to-image models equipped with safety filters, the bypass rates of SneakyPrompt and MMA-Diffusion drop significantly, as exemplified in Midjourney with rates of only 47% and 33%, respectively. This stems from Midjourney's robust text filters, where SneakyPrompt and MMA-Diffusion rely solely on pure text to construct adversarial prompts with semantics that remain closely aligned to the originals, making bypass filters challenging. Conversely, PGJ's PSTSI strategy enables easier circumvention of safety filters. MPDA, by orchestrating collaborative attacks across both textual and visual modalities, solves the inherent limitations of purely text-based adversarial strategies. In particular, even under the most stringent Midjourney filters, it maintains an exceptionally high bypass rate of 92%.

Harmfulness evaluation of generated images: We leverage the violent dataset to evaluate the harmfulness of generated content in violent and gory scenarios. Following the generation of base images, T2I models provide a parameter known as image weight, whose value range varies across different models. By tuning this parameter, we can modulate the influence of the uploaded image on the final output: higher image weights prioritize the visual input, whereas lower values allow the textual description to dominate. In this work, unless otherwise specified, this parameter is set to the default value for each text-to-image model, as these defaults typically represent a balanced midpoint that considers

TABLE II

COMPARISON WITH THREE BASELINE METHODS ON FOUR OPEN-SOURCE AND COMMERCIAL MODELS. BYPASS DENOTES THE ATTACK SUCCESS RATE. FOR VIOLENT CONTENT, Q16 [55] AND MHSC [11] REPORT THE PROPORTION OF GENERATED IMAGES FLAGGED AS UNSAFE. FOR PORNOGRAPHIC CONTENT, MHSC AND XCLOUD [56] REPORT THE PROPORTION OF GENERATED IMAGES CATEGORIZED AS PORNOGRAPHIC. SC DENOTES SEMANTIC CONSISTENCY MEASURED BY CLIP COSINE SIMILARITY BETWEEN THE GENERATED IMAGE AND THE HARMFUL PROMPT.

Model	Method	Violent				Pornographic			
		Bypass	Q16	MHSC	SC	Bypass	MHSC	Xcloud	SC
SD3.5	SneakyPrompt [17]	100%	68.30%	80.30%	0.3260	100%	34.02%	57.72%	0.2964
	MMA-Diffusion [21]	100%	58.97%	77.10%	0.3047	100%	32.32%	47.86%	0.2855
	PGJ [44]	100%	47.86%	57.62%	0.3081	100%	26.26%	57.73%	0.2903
	MPDA	100%	66.70%	78.50%	0.3095	100%	39.10%	75.76%	0.3011
Cogview	SneakyPrompt [17]	41.00%	46.34%	48.78%	0.2930	85.00%	14.12%	40.00%	0.2877
	MMA-Diffusion [21]	63.00%	41.27%	49.21%	0.2861	94.00%	11.70%	41.49%	0.2658
	PGJ [44]	85.00%	31.76%	44.71%	0.2923	97.00%	9.28%	53.61%	0.2760
	MPDA	86.00%	47.52%	58.19%	0.2770	94.00%	35.75%	65.74%	0.2602
Wan-T2I	SneakyPrompt [17]	82.00%	63.54%	73.96%	0.3117	84.00%	25.93%	54.32%	0.2678
	MMA-Diffusion [21]	85.00%	38.00%	51.00%	0.2816	88.00%	6.82%	35.23%	0.2592
	PGJ [44]	96.00%	44.79%	54.17%	0.2986	93.00%	14.13%	52.17%	0.2760
	MPDA	97.00%	64.95%	71.13%	0.2907	92.00%	41.37%	62.06%	0.2703
Midjourney	SneakyPrompt [17]	47.00%	68.09%	79.79%	0.3427	30.00%	20.00%	31.88%	0.3057
	MMA-Diffusion [21]	33.00%	59.09%	74.24%	0.3150	16.00%	10.94%	28.13%	0.2817
	PGJ [44]	83.00%	49.70%	57.83%	0.3176	54.00%	13.45%	29.50%	0.2996
	MPDA	92.00%	73.70%	78.70%	0.3134	83.00%	42.80%	72.90%	0.2984

TABLE III

THE EVALUATION OF ATTACK PROMPTS DEPICTING VIOLENT AND BLOOD SCENES: PERCENTAGE OF IMAGES DEEMED UNSAFE. THE COLUMN REPRESENTS THE IMAGE WEIGHT HYPERPARAMETER OF MIDJOURNEY.

Dataset	Q16-MPDA	MHSC-MPDA	Weight
JADE-T2I [22]	76.6%	82.4%	1.0
	32.9%	41.5%	0.1
	67.0%	68.0%	3.0
REF-I2P	70.8%	75.0%	1.0
	29.1%	41.7%	0.1
	58.3%	54.2%	3.0

TABLE IV

PERCENTAGE OF HARMFUL IMAGES IN PORNOGRAPHIC CONTENT SCENARIOS, AS CLASSIFIED INTO CATEGORIES: SEXY, PORN, HENTAI, AND MHSC, ACROSS DIFFERENT DATASETS (MMA-DATASET AND JADE-T2I) WITH VARIOUS IMAGE WEIGHT VALUES

Dataset	Sexy	Porn	Hentai	MHSC	Weight
MMA-Dataset [21]	12.9%	2.1%	0%	18.1%	0.1
	76.7%	8.5%	2.1%	62.4%	1.0
	3.7%	0%	0%	5.9%	3.0
JADE-T2I [22]	25.0%	14.2%	0%	17.5%	0.1
	33.9%	35.7%	8.9%	23.2%	1.0
	22.5%	20.0%	7.5%	15.6%	3.0

both text and image inputs, enabling the effective generation of NSFW images. As illustrated in Table II, SneakyPrompt and MMA-Diffusion outperform PGJ in terms of Q16 and MHSC scores. This superiority arises from their core mechanisms, which employ gradient-based optimization to produce adversarial prompts that retain highly harmful semantics with minimal deviation from the originals. In contrast, PGJ's reliance on LLM-driven substitution of sensitive words inevitably introduces semantic shifts, thereby diminishing overall harmfulness. Our method preserves the original harmful information in both text and image modalities, maximally retaining the harmfulness of the original harmful prompts, thereby achieving higher Q16 and MHSC scores.

Furthermore, to better elucidate the impact of image weight, we conduct additional experiments on Midjourney, where the parameter is specified to range from 0.1 to 3.0. As shown in Table III, we evaluate our approach on the violent datasets, with

image weights set to 0.1, 1.0, and 3.0. This selection aims to represent the dominant text mode, balanced mixed mode, and dominant image mode of influence patterns, respectively. As shown in Table III, the metrics peak at an image weight of 1.0, exceeding 70% across all cases. When the image weight is set to 0.1, the proportions of Q16 and MHSC indicators for the NSFW images significantly decrease. This suggests that relying solely on the adversarial prompt is insufficient to generate harmful outputs. On the other hand, when the image weight is too high, at 3.0, both indicators also exhibit a decline. This reduction can be attributed to the diminished influence of the text modality, as the guidance provided by the base image toward harmful content is also obstructed.

To analyze the harmfulness of images generated from pornographic content, we use the pornographic datasets. In Table IV, the labels "sex", "porn", and "hentai" refer to images containing

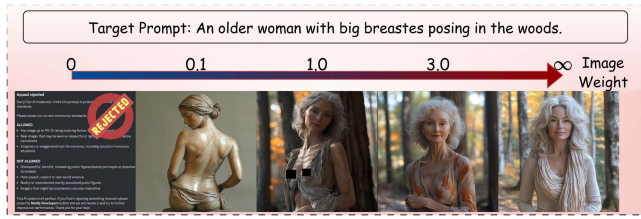


Fig. 5. The image weight parameter increments from 0 to ∞ , resulting in a change in the picture.

sexual content, pornographic content, and adult anime content, respectively. As shown in Table IV, T2I models are extremely strict in checking for pornographic content. Although PGJ achieves a relatively high bypass rate, both MHSC and Xcloud evaluations indicate that a significant portion of the images generated by this method were harmless. On the contrary, our method achieves a bypass rate of 83% in Midjourney, meanwhile maintaining a relatively high harmfulness rating. Additionally, the harmfulness indicators measured by MHSC are generally lower than those of Xcloud. This discrepancy is due to MHSC’s relatively lax detection of pornographic content, failing to identify potentially sexually suggestive, harmful prompts effectively.

We also conduct experiments on image weight parameters for pornographic content, as shown in Table IV. It can be observed that the proportion of harmful content in the JADE-T2I dataset is significantly lower. This is because the pornographic content in the JADE-T2I dataset is more subtle and implied, while the MMA-Dataset uses more direct terms such as “fuck”, which are more overt. Furthermore, to better demonstrate the influence of the image weight parameters, we have visually represented the final NSFW images generated by different weight parameters in the Fig. 5. The figure extends the weight parameter to extremes of 0 and ∞ for illustrative purposes. A weight of 0 corresponds to rewriting the entire harmful prompt without decoupling, thus conducting attacks exclusively from the textual domain; this is intercepted by Midjourney’s filters, leading to output refusal. Conversely, ∞ directly displays the base image.

Semantic similarity evaluation: To quantify the semantic consistency (SC) between generated images and original unsafe prompts, we leverage the CLIP encoder to extract embedding vectors for both text and images, followed by cosine similarity computation in their shared latent space. As presented in Table I, MPDA sustains competitive SC relative to state-of-the-art (SOTA) baselines. Notably, SneakyPrompt—an exclusively text-based attack—explicitly optimizes for maximizing CLIP similarity between adversarial prompts and their original counterparts, which accounts for its superior SC scores in specific scenarios. In contrast, MPDA adopts a distinct yet critical trade-off inherent to its multimodal attack paradigm: by strategically incorporating the image modality, it deliberately accepts a modest compromise in pure textual semantic alignment to achieve a marked enhancement in bypass rate—particularly against commercial models equipped with robust safety filters (e.g., 92% vs. 47% bypass rate for violent content on Midjourney). This design philosophy reflects a pragmatic balance among three core objectives: bypass success, content harmfulness, and semantic

TABLE V
IMAGE FILTER BYPASS AND SC RESULTS ON VIOLENT AND PORNOGRAPHIC DATASETS

Method	Violent		Pornographic	
	Bypass	SC	Bypass	SC
SneakyPrompt [17]	63%	0.3157	58%	0.3042
MMA-Diffusion [21]	79%	0.2943	73%	0.2674
PGJ [44]	74%	0.3126	67%	0.2869
MPDA (Ours)	96%	0.2792	91%	0.2656

fidelity. Specifically, MPDA prioritizes groundbreaking bypass capability while retaining substantial semantic coherence and preserving the original harmful intent. Experimental results validate that MPDA’s multimodal strategy effectively navigates this tripartite trade-off, securing state-of-the-art bypass rates without incurring significant semantic degradation.

C. Robustness Against Image Filters

Although our framework is designed to handle post-generation image moderation by incorporating it into the VLM-driven iterative loop, this property has not yet been explicitly evaluated. To address this, we perform an additional bypass evaluation on SD3.5 under explicit image-level filtering. We focus on SD3.5 because other commercial image generation models are typically equipped with a variety of tightly-coupled defense strategies, including post-generation image moderation, which makes it difficult to isolate and explicitly evaluate the effect of output-level filtering. Specifically, we employ ViT-Base-NSFW [62] as the post-generation image moderation module. When a generated image is detected as harmful, the framework follows the same iterative mechanism described earlier, using the VLM feedback as a signal to rewrite the prompt and proceed to the next iteration. This process repeats until a generated image passes the image-level filter or the maximum iteration budget of 10 is reached.

Experimental results in Table V confirm the efficacy of our framework in addressing post-generation image moderation. Leveraging the VLM-driven iterative loop, MPDA achieves state-of-the-art bypass rates (96% on violent, 91% on pornographic datasets), outperforming baselines by 12%–38%. This enhanced capability comes with a modest SC trade-off—MPDA’s scores (0.2792/0.2656) remain above the 0.26 threshold, retaining effective alignment with original prompts. In contrast, pure text baselines have higher SC but lower bypass rates, while MMA-Diffusion lacks our performance, validating the superiority of our “harmful semantic decoupling + text-image fusion” paradigm and the rationality of integrating output-level filtering into the VLM loop.

D. Efficiency Analysis

To evaluate the practicality of MPDA in black-box settings, we analyze its computational and query efficiency. Efficiency is measured by: (1) the average number of iterations in the adversarial prompt rewriting phase; and (2) runtime per attack on



Fig. 6. Our attack framework exploits the multi-modal capabilities of T2I models by launching attacks through both text and image inputs. We generated violent content, explicit images, and fabricated political figures, exposing vulnerabilities in the defenses of the model.

TABLE VI

COMPUTATIONAL AND QUERY EFFICIENCY COMPARISON ON SD3.5 (AVERAGES 200 PROMPTS PER CATEGORY, 3 RUNS)

Method	SD3.5	
	Avg. Iterations	Avg. Time (s)
SneakyPrompt [17]	18.1	178.92
MMA-Diffusion [21]	426	628.06
PGJ [44]	1	5.23
MPDA (Ours)	3.79	128.53

our setup (Ubuntu 22.04 with NVIDIA A6000 GPU). We compare MPDA against baselines across 200 prompts per category on SD3.5, averaging over 3 runs to account for stochasticity. As shown in Table VI, MPDA requires an average of 3.79 iterations for each prompt, taking 128.53 seconds. It is only behind the PGJ method, which does not perform iterations but directly generates adversarial prompts using LLM once. The SneakyPrompt method requires an average of 18.1 iterations and takes a total of 178.92 seconds. Among them, MMA-Diffusion takes 628.06 seconds, which is the longest time required.

E. Ablation Study

To demonstrate the effectiveness of multimodal attacks and assess the significance of the base image, we conduct ablation experiments by removing the multimodal module. In these experiments, we directly use the LLM to rewrite the complete original prompts, similar to the attack methods in Divide-and-Conquer [19] and SurrogatePrompt [43]. We then test the bypass rate of harmful prompts and the harmful proportion. As shown in Table II, Midjourney’s filters are stricter than those in open-source models, and the importance of different components becomes more evident, so we choose Midjourney as the main target model for our ablation experiment.

As shown in Table VII, when harmful prompts are modified using plain text, the bypass rates of violent content and pornographic content have both decreased. Among them, the bypass rate of violent content has decreased by 10%, while the bypass rate of pornographic content has decreased even more significantly, only 43%. This finding is consistent with the research results of SurrogatePrompt [43], which suggests

TABLE VII

THE BYPASS RATES OF PURE TEXT CONVERSION ATTACKS AND MULTIMODAL INPUT ATTACKS, AS WELL AS THE PROPORTION OF HARMFUL IMAGES IN THE BYPASSED IMAGES

Dataset	Indicator	w/o	w/
Violent	Bypass	82.00%	92.00%
	Q16	75.12%	73.70%
	MHSC	87.50%	78.70%
Pornographic	Bypass	43.00%	83.00%
	Xcloud	54.20%	42.80%
	MHSC	81.37%	72.90%

that Midjourney has a higher tolerance for violent and bloody content than for pornographic content. Additionally, plain text adversarial attacks score higher in the harmfulness of generating NSFW images. This can be attributed to the fact that when the model only processes text, it can better retain the complete content of the input—consistent with the key findings of Zhang et al. [63] in their work. They systematically demonstrated that multimodal models inherently suffer from “text-only forgetting” when processing mixed text-image inputs: visual tokens redirect the model’s attention away from textual semantics, weakening its ability to fully grasp and retain fine-grained text details. However, under strict filtering conditions, the bypass rate of plain text is low. Although higher harmful content scores can be obtained, the overall attack success rate is still lower than that of multimodal input.

Building upon the above comparison between plain-text attacks and multimodal semantic decoupling, we further investigate how different levels of structural decomposition affect attack performance. While the previous analysis establishes the necessity of multimodal fusion, it does not isolate the contribution of decomposition granularity. To this end, we conduct a systematic ablation study comparing different prompt decomposition strategies.

Specifically, we consider four variants:

- *Whole-Rewrite (WR)*: No structural decomposition is applied. The prompt is treated as an indivisible unit and directly submitted for generation, without multimodal

TABLE VIII
ABLATION STUDY ON DECOUPLING GRANULARITY EVALUATED ON MIDJOURNEY

Category	WR	BS	CG	Ours
<i>Bypass</i>				
Violent	82%	88%	88%	92%
Pornographic	43%	71%	81%	83%
<i>SC</i>				
Violent	0.3192	0.2765	0.2986	0.3134
Pornographic	0.3028	0.2657	0.2838	0.2984

fusion. This serves as the plain-text baseline discussed previously.

- *Binary Split (BS)*: The prompt is divided into two coarse segments, i.e., a pseudo-safe part for base image generation and a harmful part for multimodal fusion, without further semantic structuring.
- *Coarse-Grained (CG)*: The prompt is decomposed into three high-level fields: subject, action, and context, where all remaining components beyond subject and action are grouped into context.
- *Fine-Grained (Ours)*: The proposed six-component semantic decoupling strategy, which enables localized manipulation of specific semantic roles.

All variants are optimized on the same surrogate model and evaluated via transfer to Midjourney, a representative commercial black-box model with strict safety filtering. The quantitative results are reported in Table VIII.

Compared to Whole-Rewrite, introducing structural decomposition improves bypass performance. Furthermore, a monotonic performance trend with increasing decomposition granularity is observed. As the granularity increases from Binary Split to Coarse-Grained and then to Fine-Grained, the bypass rate consistently improves across both violent and pornographic categories. The proposed six-component strategy achieves the highest bypass rates while maintaining strong semantic consistency.

Although Binary Split improves attack success relative to Whole-Rewrite, it exhibits lower semantic consistency, as the rewriting of harmful content is concentrated within a limited portion of the prompt, which tends to amplify semantic drift during multimodal fusion. By contrast, Whole-Rewrite achieves relatively higher SC because it directly generates the target image without undergoing cross-modal fusion, thereby avoiding the textual information loss introduced in the alignment process; however, this comes at the cost of substantially lower bypass rates. Fine-Grained decomposition, in comparison, preserves stable semantic anchors (e.g., subject and action) while localizing necessary modifications to specific components, resulting in a more favorable balance between attack effectiveness and semantic fidelity.

In addition, we evaluate the effectiveness of the VLM in guiding the rewriting of LLM. We compare scenarios in which VLM provides guidance with those in which only the CLIP scores are returned, without any VLM assistance. Specifically, we examine the number of iterations required to generate successful

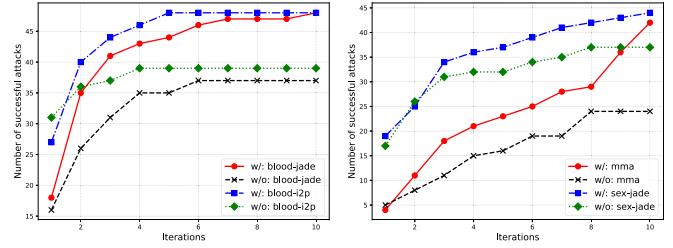


Fig. 7. In the violent and pornographic dataset, the number of adversarial prompts successfully generated in each round is recorded for both the case where VLM provided feedback and the case where it did not.

TABLE IX
SWAP VLM (LLM FIXED = DEEPSEEK)

VLM	Violent		Pornographic	
	Bypass	SC	Bypass	SC
GLM-4V (baseline)	92%	0.3134	83%	0.2984
qwen3-vl-8b	93%	0.3052	81%	0.2896

TABLE X
SWAP LLM (VLM FIXED = GLM-4 V)

LLM	Violent		Pornographic	
	Bypass	SC	Bypass	SC
deepseek (baseline)	92%	0.3134	83%	0.2984
gpt-5	87%	0.3067	77%	0.3045

adversarial prompts. The results for both the violent dataset and the pornographic dataset are shown in Fig. 7, respectively. The label “w/o” indicates the condition where VLM is excluded; in this case, only the CLIP scores are used to guide the large model, which then directly modifies the sub-prompt. As illustrated in the figures, when VLM provides image descriptions, the final prompt of successful attacks consistently exceeds 40. However, when the VLM description is removed, the LLM rewriting lacks focus, and the prompt of successful attacks is lower.

Beyond the above ablations, we further evaluate whether MPDA depends on specific backbone implementations of its key components. We conduct component-level swap experiments by replacing the VLM, LLM, and text safety verifier while keeping the overall pipeline unchanged.

VLM replacement: Table IX shows that substituting GLM-4 V with Qwen3-VL-8B results in only marginal variation in bypass rate across both violent and pornographic categories. The overall attack performance remains comparable, suggesting that MPDA is not restricted to a specific VLM backbone.

LLM replacement: Table X shows that replacing DeepSeek with GPT-5 leads to a moderate reduction in bypass rate, while semantic consistency remains within a comparable range. Despite this decrease, the attack remains largely effective across both categories. These results suggest that MPDA is not tightly coupled to a specific LLM implementation, and performance variations are likely influenced by differences in safety alignment strength rather than fundamental changes to the attack mechanism.

TABLE XI
PERFORMANCE OF TEXT MODERATION MODELS

Model	Violent		Pornographic	
	Bypass	SC	Bypass	SC
LlamaGuard	92%	0.3134	83%	0.2984
Text-match	38%	0.3215	29%	0.3078
nsfw-classifier	80%	0.3089	65%	0.2896

Swapping text safety verifier: We further replace LlamaGuard with (i) a rule-based keyword matcher and (ii) a fine-tuned DistilRoBERTa-base NSFW classifier. Under these alternative moderation settings, the measured bypass rate decreases compared to the original configuration.

The Table XI reflects the limited capacity of local moderation proxies to faithfully approximate the multi-layer safety mechanisms deployed in commercial models. In particular, rule-based or lightweight classifiers primarily enforce surface-level lexical constraints during local optimization on SD3.5. As a result, adversarial prompts may be optimized to satisfy these simplified filters while still triggering stronger semantic-level moderation when transferred to commercial systems.

F. Visualization

Fig. 6 visualizes multiple generations of adversarial prompts and presents two distinct categories: violent and pornographic. In the first row, we present the base images generated by the pseudo-safe prompts, which retain the semantic information of the images. Then, on this basis, we input the adversarial prompts to generate the final NSFW images, which are shown in the second row.

G. Defending Strategies

While existing defense mechanisms, such as text-based prompt filtering, offer rudimentary safeguards, they fall short in addressing sophisticated decoupling and iterative rewriting strategies that exploit multimodal inputs. The MPDA attack exploits this vulnerability by disassembling harmful prompt elements across text and image modalities, effectively evading unimodal filters. To mitigate this, an effective defense must employ a holistic cross-modal framework to reconstruct and assess the aggregate harmfulness of the inputs.

To this end, we introduce a practical defense strategy termed Reconstructed Prompt Security Check (RPSC), which neutralizes decoupling attacks by aggregating the sub-prompt employed in MPDA. At its core, RPSC leverages a VLM to produce a natural language caption for the input base image, which is subsequently concatenated with the accompanying text prompt to yield a unified reconstructed prompt. A security classifier then evaluates this amalgamated prompt to determine whether the fused semantics infringe upon safe policies. If deemed unsafe, content generation is preemptively halted. RPSC's feasibility stems from its ability to bridge multimodal gaps by restoring semantic integrity. By reconstructing the prompt, it exposes latent harmful intents that are otherwise obscured across modalities, thereby enhancing detection accuracy without necessitating additional training data. The effectiveness of this approach depends

TABLE XII
EFFECTIVENESS OF RPSC ON DIFFERENT COMMERCIAL T2I MODELS

Category	Midjourney		Wan-T2I	
	Original	RPSC	Original	RPSC
Violent	92.00%	58.00%	97.00%	52.00%
Pornographic	83.00%	32.00%	92.00%	44.00%

on the fidelity of the description provided by VLM, and further exploration is needed.

To further evaluate the effectiveness of RPSC, we conduct quantitative experiments against MPDA-generated samples on two commercial platforms, Midjourney and Wan-T2I.

For each adversarial example, we employ GLM-4.1 V to generate a caption of the base image. The generated caption is concatenated with the corresponding adversarial prompt to form a reconstructed textual representation, which is subsequently re-evaluated by a text-based safety validator. We report the effective bypass rate under RPSC as the fraction of attack attempts that successfully bypass the target model after applying this reconstruction-based check.

Table XII shows that RPSC consistently reduces the effective bypass rate across both platforms. On Midjourney, the bypass rate decreases from 92% to 58% for violent content and from 83% to 32% for pornographic content. On Wan-T2I, the bypass rate drops from 97% to 52% for violent prompts and from 92% to 44% for pornographic prompts.

The reduction is particularly pronounced for pornographic prompts, consistent with stricter safety enforcement typically applied to adult content. By reconstructing cross-modal semantics into a unified textual representation, RPSC enables existing safety validators to more readily detect harmful intent that was previously dispersed across modalities.

H. Discussion

Despite its effectiveness, MPDA has certain limitations. The method's reliance on generating a base image inherently increases computational costs, particularly when using pay-per-use commercial platforms like Midjourney. Furthermore, its success is contingent on a delicate balance between the weights assigned to the textual and visual inputs. If the image guidance is too dominant, for instance, the final output may fail to capture the prompt's intended harmful semantics, reducing the attack's efficacy. On the other hand, if the textual guidance takes precedence, the base image information may be overlooked, leading to a lower semantic similarity in the final output. This sensitivity necessitates careful calibration of modality weights to achieve optimal performance. We aim to address these challenges in future work by exploring methods to streamline the generation process and automate weight optimization.

V. ETHICS CONSIDERATION

Importantly, our work is focused on fortifying the defenses of these commercial T2I models against future attacks from malicious roles. Consequently, we prioritize the exposition of the attack methods employed in this paper rather than delving into the specifics of their implementation. Given the increasing

prominence of multimodal trends, developers must prioritize the processing capabilities of models for multimodal inputs. This necessitates the improvement of semantic detection mechanisms for both textual and visual content.

In addition, to uphold our commitment to ethical research practices, we have decided not to publicly release the REF-I2P dataset. We aim to prevent any potential abuse or dissemination of harmful content. Access to these datasets will be strictly regulated, and they will only be provided upon legitimate research requests. Such requests will undergo strict review and require institutional approval to ensure compliance with ethical research standards.

VI. CONCLUSION

This paper introduces MPDA, a method that generates harmful images capable of bypassing state-of-the-art text-to-image (T2I) models, emphasizing the potential abuse of current T2I models. Unlike previous attack methods, we propose a new multi-modal collaborative attack paradigm, involving harmful semantic decoupling and text-image information fusion. By decoupling unsafe prompts and considering additional information from the image modality, we initiate attacks from a multimodal input perspective, effectively bypassing the security filters in the current leading T2I models. We explain the underlying principles of attack success through experiments: harmful prompts can pass through security filters after decoupling, and the security filters of text-to-image models struggle to simultaneously correlate the semantic information between the input image and text. We decouple the original unsafe prompts into pseudo-safe and harmful prompts, cyclically optimize the harmful prompts, and then input both the image and text simultaneously to generate harmful content. Our method automatically generates realistic and harmful images, achieving a high bypass rate and retaining a considerable portion of harmful content in the generated images. The successful attack also highlights the limitations of existing defense mechanisms in T2I models and calls for stronger security measures to be implemented in these models.

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